



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment-9

Experiment No.: 9

Title: Implement localStorage and sessionStorage to save user preferences like theme settings.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code, Google Chrome

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Preference Studio</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="main">
    <section class="intro">
      <p class="label">
        SHRUTI BAWANKAR
      </p>
      <h1>
        Make it feel<br>
        like yours.
      </h1>
      <p class="description">
        Choose your theme and where to store it.
        PRN : 24070521148
      </p>
      <button
        id="resetLink"
        class="reset-link">
        Reset all settings
      </button>
      <div class="current-card">
        <div class="current-header">
          <div class="current-icon">
            ⚙️
          </div>
          <div>
            <h2>
              Current Preference
            </h2>
          </div>
        </div>
      </div>
    </section>
  </main>
</body>
</html>
```



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```
</h2>
<p>
  Here's what's currently applied.
</p>
</div>
</div>
<div class="current-row">
  <div class="row-icon theme-row-icon">
    ●
  </div>
  <span>
    Theme
  </span>
  <strong id="currentTheme">
    Not selected
  </strong>
</div>
<div class="current-row">
  <div class="row-icon storage-row-icon">
    ☰
  </div>
  <span>
    Storage Method
  </span>
  <strong id="currentStorage">
    Not selected
  </strong>
</div>
</div>
</section>
<section class="settings-card">
  <div class="setting-section">
    <div class="step-number">
      01
    </div>
    <div class="setting-content">
      <h2>
        Choose a theme
      </h2>
      <p class="hint">
        Select the appearance you prefer.
      </p>
      <div class="choice-grid">
```



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```
<label class="choice">
  <input
    type="radio"
    name="theme"
    value="light"
  >
  <div class="choice-icon light-icon">
    ☀
  </div>
  <div class="choice-text">
    <strong>
      Light
    </strong>
    <span>
      Bright and clear
    </span>
  </div>
  <span class="radio"></span>
</label>
<label class="choice">
  <input
    type="radio"
    name="theme"
    value="dark"
  >
  <div class="choice-icon dark-icon">
    ☾
  </div>
  <div class="choice-text">
    <strong>
      Dark
    </strong>
    <span>
      Easy on the eyes
    </span>
  </div>
  <span class="radio"></span>
</label>
</div>
</div>
<div class="setting-section">
  <div class="step-number">
    02
```



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```
</div>
<div class="setting-content">
  <h2>
    Save preference using
  </h2>
  <p class="hint">
    Choose where to store your theme setting.
  </p>
  <div class="choice-grid">
    <label class="choice">
      <input
        type="radio"
        name="storage"
        value="local"
      >
      <div class="choice-icon storage-icon">
        ☐
      </div>
      <div class="choice-text">
        <strong>
          Local Storage
        </strong>
        <span>
          Stays even after you close
          the browser
        </span>
      </div>
      <span class="radio"></span>
    </label>
    <label class="choice">
      <input
        type="radio"
        name="storage"
        value="session"
      >
      <div class="choice-icon session-icon">
        ☐
      </div>
      <div class="choice-text">
        <strong>
          Session Storage
        </strong>
        <span>
          Stays only for the current
```



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```
        browser session
    </span>
</div>
    <span class="radio"></span>
</label>
</div>
</div>
<div class="setting-section last-section">
    <div class="step-number">
        03
    </div>
    <div class="setting-content">
        <h2>
            Save or clear
        </h2>
        <p class="hint">
            Apply your selected preference or remove it.
        </p>
        <div class="action-buttons">
            <button
                id="saveBtn"
                class="save-btn">
                Save Preference
            </button>
            <button
                id="clearBtn"
                class="clear-btn">
                Clear Preference
            </button>
        </div>
        <p
            id="message"
            class="message">
        </p>
    </div>
</div>
</section>
</main>
<script src="script.js"></script>
</body>
</html>
```



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Script.js:

```
const saveBtn =  
    document.getElementById("saveBtn");  
const clearBtn =  
    document.getElementById("clearBtn");  
const resetLink =  
    document.getElementById("resetLink");  
const message =  
    document.getElementById("message");  
const currentTheme =  
    document.getElementById("currentTheme");  
const currentStorage =  
    document.getElementById("currentStorage");  
const statusText =  
    document.getElementById("statusText");  
function applyTheme(theme) {  
    if (theme === "dark") {  
        document.body.classList.add("dark");  
    } else {  
        document.body.classList.remove("dark");  
    }  
}  
saveBtn.addEventListener("click", function () {  
    const theme =  
        document.querySelector(  
            'input[name="theme"]:checked'  
        );  
    const storage =  
        document.querySelector(  

```



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```
'input[name="storage"]:checked'

);

if (!theme || !storage) {
  message.textContent =
    "Please select a theme and storage method.";
  message.style.color = "#d9534f";
  return;
}

const selectedTheme =
  theme.value;

const selectedStorage =
  storage.value;

if (selectedStorage === "local") {
  localStorage.setItem(
    "theme",
    selectedTheme
  );
};

sessionStorage.removeItem("theme");

currentStorage.textContent =
  "Local Storage";
}

else {
  sessionStorage.setItem(
    "theme",
    selectedTheme
  );
};

localStorage.removeItem("theme");

currentStorage.textContent =
```



```
"Session Storage";
}
applyTheme(selectedTheme);
if (selectedTheme === "dark") {
    currentTheme.textContent = "Dark";
} else {
    currentTheme.textContent = "Light";
}
message.textContent =
    "Preference saved successfully!";
message.style.color = "#45a66d";
statusText.textContent =
    "Preferences saved";
});
function loadPreference() {
    let savedTheme = null;
    let savedStorage = null;
    const localTheme =
        localStorage.getItem("theme");
    const sessionTheme =
        sessionStorage.getItem("theme");
    if (localTheme !== null) {
        savedTheme = localTheme;
        savedStorage = "Local Storage";
    }
    else if (sessionTheme !== null) {
        savedTheme = sessionTheme;
        savedStorage = "Session Storage";
```



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```
}  
if (savedTheme !== null) {  
    applyTheme(savedTheme);  
    currentTheme.textContent =  
        savedTheme === "dark"  
        ? "Dark"  
        : "Light";  
    currentStorage.textContent =  
        savedStorage;  
    document.querySelector(  
        `input[name="theme"][value="${savedTheme}"]`  
    ).checked = true;  
    const storageValue =  
        savedStorage === "Local Storage"  
        ? "local"  
        : "session";  
    document.querySelector(  
        `input[name="storage"][value="${storageValue}"]`  
    ).checked = true;  
}  
}  
  
function clearPreference() {  
    localStorage.removeItem("theme");  
    sessionStorage.removeItem("theme");  
    applyTheme("light");  
    currentTheme.textContent =  
        "Not selected";  
    currentStorage.textContent =
```



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```
"Not selected";

document.querySelectorAll(
    'input[name="theme"]'
).forEach(function (radio) {
    radio.checked = false;
});

document.querySelectorAll(
    'input[name="storage"]'
).forEach(function (radio) {
    radio.checked = false;
});

message.textContent =
    "Preference cleared.";
message.style.color = "#45a66d";
statusText.textContent =
    "Preferences reset";
}

clearBtn.addEventListener(
    "click",
    clearPreference
);

resetLink.addEventListener(
    "click",
    clearPreference
);

loadPreference();
```



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Style.css:

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
body {
  font-family: Arial, sans-serif;
  background: #f7f3ed;
  color: #252525;
  min-height: 100vh;
  padding: 0 4%;
  transition: 0.3s;
}
.main {
  max-width: 1250px;
  margin: 70px auto;
  display: grid;
  grid-template-columns: 0.9fr 1.1fr;
  gap: 55px;
  align-items: start;
}
.intro {
  padding-top: 10px;
}
.label {
  color: #d85f3d;
  font-size: 13px;
  font-weight: bold;
  letter-spacing: 3px;
  margin-bottom: 28px;
}
.intro h1 {
  font-size: clamp(48px, 5vw, 72px);
  line-height: 1.05;
  letter-spacing: -3px;
  margin-bottom: 25px;
}
.description {
  color: #737373;
  font-size: 20px;
  line-height: 1.6;
  max-width: 500px;
```



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Experiment-9

```
margin-bottom: 28px;
}
.reset-link {
  border: none;
  border-bottom: 1px solid #777;
  background: transparent;
  color: #666;
  padding-bottom: 6px;
  font-size: 15px;
  cursor: pointer;
}
.reset-link:hover {
  color: #d85f3d;
  border-color: #d85f3d;
}
.settings-card {
  background: #fffdfa;
  border: 1px solid #ded9d2;
  border-radius: 15px;
  padding: 30px 35px;
  box-shadow:
    0 15px 40px rgba(60, 50, 40, 0.06);
}
.setting-section {
  display: grid;
  grid-template-columns: 45px 1fr;
  gap: 10px;
  padding-bottom: 28px;
  margin-bottom: 28px;
  border-bottom: 1px solid #e3ded7;
}
.last-section {
  border-bottom: none;
  margin-bottom: 0;
  padding-bottom: 0;
}
.step-number {
  color: #e66d4b;
  font-size: 20px;
  font-weight: 500;
}
.setting-content h2 {
  font-size: 22px;
  margin-bottom: 5px;
}
```



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Experiment-9

```
}  
.hint {  
  color: #777;  
  font-size: 15px;  
  margin-bottom: 18px;  
}  
.choice-grid {  
  display: grid;  
  grid-template-columns: 1fr 1fr;  
  gap: 15px;  
}  
.choice {  
  position: relative;  
  display: flex;  
  align-items: center;  
  gap: 14px;  
  min-height: 95px;  
  padding: 16px;  
  border: 1px solid #e2e0dc;  
  border-radius: 11px;  
  background: #f7f6f4;  
  cursor: pointer;  
  transition: 0.2s;  
}  
.choice:hover {  
  border-color: #e77857;  
}  
.choice input {  
  display: none;  
}  
.choice:has(input:checked) {  
  border: 2px solid #e87554;  
  background: #fffaf7;  
}  
.choice-icon {  
  min-width: 52px;  
  width: 52px;  
  height: 52px;  
  display: flex;  
  align-items: center;  
  justify-content: center;  
  border-radius: 50%;  
  font-size: 27px;  
}
```



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```
.light-icon {
  background: #fde8d9;
  color: #d87539;
}
.dark-icon {
  background: #e3e5e6;
  color: #182431;
}
.storage-icon {
  background: #fde6d8;
  color: #d65e3b;
}
.session-icon {
  background: #dfeaf7;
  color: #31658f;
}
.choice-text {
  display: flex;
  flex-direction: column;
  gap: 5px;
  padding-right: 20px;
}
.choice-text strong {
  font-size: 15px;
}
.choice-text span {
  color: #777;
  font-size: 13px;
  line-height: 1.4;
}
.radio {
  position: absolute;
  top: 18px;
  right: 16px;
  width: 22px;
  height: 22px;
  border: 2px solid #bbb;
  border-radius: 50%;
}
.choice:has(input:checked) .radio {
  border: 2px solid #e87554;
}
.choice:has(input:checked) .radio::after {
  content: "";
```



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```
position: absolute;
width: 10px;
height: 10px;
background: #e87554;
border-radius: 50%;
top: 4px;
left: 4px;
}
.action-buttons {
display: flex;
gap: 15px;
}
.action-buttons button {
padding: 13px 27px;
border-radius: 8px;
font-size: 15px;
cursor: pointer;
}
.save-btn {
border: none;
background: #df704f;
color: white;
}
.save-btn:hover {
background: #c95d3d;
}
.clear-btn {
border: 1px solid #d7d2cc;
background: #fffdfa;
color: #333;
}
.clear-btn:hover {
background: #f3f0ec;
}
.message {
margin-top: 12px;
font-size: 13px;
color: #45a66d;
}
.current-card {
margin-top: 35px;
background: #fffdfa;
border: 1px solid #ded9d2;
border-radius: 15px;
```



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```
padding: 25px 28px;
box-shadow:
  0 12px 35px rgba(60, 50, 40, 0.05);
}
.current-header {
  display: flex;
  align-items: center;
  gap: 14px;
  margin-bottom: 18px;
}
.current-icon {
  width: 45px;
  height: 45px;
  display: flex;
  align-items: center;
  justify-content: center;
  background: #f0efec;
  border-radius: 50%;
  color: #666;
}
.current-header h2 {
  font-size: 20px;
  margin-bottom: 4px;
}
.current-header p {
  color: #777;
  font-size: 13px;
}
.current-row {
  display: flex;
  align-items: center;
  gap: 14px;
  background: #f5f5f3;
  border: 1px solid #e8e6e2;
  border-radius: 10px;
  padding: 12px 16px;
  margin-top: 10px;
  font-size: 14px;
}
.current-row span {
  color: #777;
}
.current-row strong {
  margin-left: auto;
```



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```
}  
.row-icon {  
  width: 32px;  
  height: 32px;  
  display: flex;  
  align-items: center;  
  justify-content: center;  
  border-radius: 50%;  
}  
.theme-row-icon {  
  background: #fde6d9;  
  color: #d56845;  
}  
.storage-row-icon {  
  background: #dce9f7;  
  color: #356b98;  
}  
body.dark {  
  background: #1d2025;  
  color: #f5f5f5;  
}  
body.dark .header {  
  border-color: #383b40;  
}  
body.dark .description {  
  color: #aaa;  
}  
body.dark .status {  
  color: #aaa;  
}  
body.dark .settings-card,  
body.dark .current-card {  
  background: #292c31;  
  border-color: #41444a;  
}  
body.dark .setting-section {  
  border-color: #41444a;  
}  
body.dark .choice {  
  background: #32353a;  
  border-color: #484b50;  
}  
body.dark .choice:has(input:checked) {  
  background: #38343a;
```



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```
border-color: #e87554;
}
body.dark .choice-text span {
  color: #aaa;
}
body.dark .clear-btn {
  background: #292c31;
  color: white;
  border-color: #555;
}
body.dark .current-row {
  background: #32353a;
  border-color: #484b50;
}
body.dark .current-header p {
  color: #aaa;
}
@media (max-width: 900px) {
  .main {
    grid-template-columns: 1fr;
    gap: 35px;
    margin-top: 45px;
  }
  .intro {
    padding-top: 0;
  }
}
@media (max-width: 600px) {
  body {
    padding: 0 20px;
  }
  .header {
    height: 80px;
  }
  .logo {
    font-size: 17px;
  }
  .logo-icon {
    width: 38px;
    height: 38px;
  }
  .status {
    font-size: 12px;
  }
}
```



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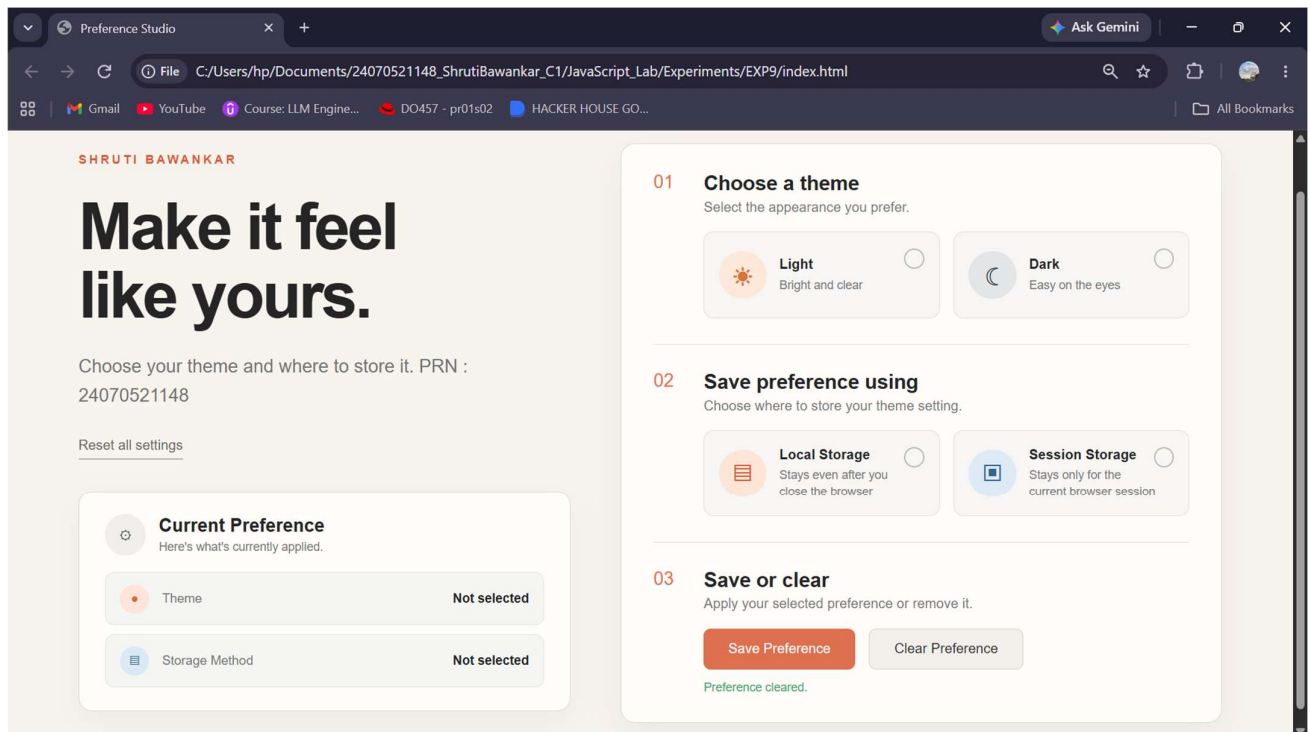
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Experiment-9

```
.intro h1 {
  font-size: 48px;
  letter-spacing: -2px;
}
.description {
  font-size: 17px;
}
.settings-card {
  padding: 22px 18px;
}
.choice-grid {
  grid-template-columns: 1 fr;
}
.action-buttons {
  flex-direction: column;
}
.action-buttons button {
  width: 100%;
}
}
```

Output:





SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment-9

SHRUTI BAWANKAR

Make it feel like yours.

Choose your theme and where to store it. PRN : 24070521148

[Reset all settings](#)

Current Preference
Here's what's currently applied.

Theme: **Dark**

Storage Method: **Local Storage**

01 Choose a theme
Select the appearance you prefer.

Light Bright and clear

Dark Easy on the eyes

02 Save preference using
Choose where to store your theme setting.

Local Storage Stays even after you close the browser

Session Storage Stays only for the current browser session

03 Save or clear
Apply your selected preference or remove it.

Save Preference **Clear Preference**

Preference saved successfully!

Result & Conclusion: The webpage successfully stores and retrieves the selected theme using either localStorage or sessionStorage. Thus, the practical demonstrates how browser storage can be used to save and manage user preferences effectively.